

Machine Learning Techniques for Reconstructing the Assembly History of the Milky Way

by

Andrea Sante

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Doctor of Philosophy

Month Year

Declaration

The work presented in this thesis was carried out at the Astrophysics Research Institute, Liverpool John Moores University. Unless otherwise stated, it is the original work of the author.

While registered as a candidate for the degree of Doctor of Philosophy, for which submission is now made, the author has not been registered as a candidate for any other award. This thesis has not been submitted in whole, or in part, for any other degree.

Andrea Sante

Astrophysics Research Institute
Liverpool John Moores University
IC2, Liverpool Science Park
146 Brownlow Hill
Liverpool
L3 5RF
UK

Abstract

[Abstract goes here.]

In this thesis I develop data-driven and machine-learning methodologies that improve upon those traditionally adopted for the reconstruction of the Milky Way’s assembly history. Throughout, I exploit cosmological hydrodynamical zoom-in simulations of Milky Way-mass haloes—in which the origin of every star particle is known—to build, validate and calibrate each method before applying it to observations. The work follows the logical arc of the standard workflow and extends it end to end: classifying accreted versus in-situ stars (Chapter 2), grouping accreted debris into progenitors (Chapter 3), characterising the properties of those progenitors at accretion (Chapter 4), and finally applying this machinery to the real Galaxy (Chapter 5).

Publications

During the course of the preparation of this thesis, the work within Chapters 2, 3, 4, and 5 has been presented in the following jointly authored publications:

1 ‘*Applying machine learning to Galactic Archaeology: how well can we recover the origin of stars in Milky Way-like galaxies?*’

Sante, A., Font, A. S., Ortega-Martorell, S., Olier, I., & McCarthy, I. G.

2 ‘*Optimized HDBSCAN clustering for reconstructing the merger history of the Milky Way: applications and limitations*’

Sante, A., Font, A. S., Mistry, I., Ortega-Martorell, S., & Olier, I.

3 ‘*GalactiKit: reconstructing mergers from $z = 0$ debris using simulation-based inference in Auriga*’

Sante, A., Kawata, D., Font, A. S., & Grand, R. J. J.

4 ‘*A simulation-based inference of the Milky Way merger history*’

Sante, A., Font, A. S., Kawata, D., Makinen, T. L., & Grand, R. J. J.

Publication 1 forms the basis of Chapter 2. It presents several machine learning models trained on the ARTEMIS cosmological zoom-in simulations to separate accreted from in-situ stars in Milky Way-mass galaxies. All analysis was carried out by A. Sante with major scientific contributions from A.S. Font. The implementation of the machine learning models was reviewed and supported by

Publication 2 forms the basis of Chapter 3. It investigates the effectiveness of the HDBSCAN clustering algorithm for recovering the progenitors of Milky Way-type galaxies using the Auriga simulations, introducing a 12-dimensional chemodynamical feature space and an optimisation strategy to reduce fragmentation. All analysis was carried out by A. Sante.

Publication 3 forms the basis of Chapter 4. It presents GalactiKit, a simulation-based inference framework for estimating the infall time, stellar mass, halo mass and mass ratio of disrupted progenitors from the properties of their $z = 0$ debris in the Auriga simulations. All analysis was carried out by A. Sante.

Publication 4 forms the basis of Chapter 5. It applies a normalizing-flow simulation-based inference framework, trained on Auriga debris samples, to a local sample of accreted stars in the Milky Way, inferring the accretion times and masses of several known merger events and placing constraints on the growth of the Galactic halo. All analysis was carried out by A. Sante.

Acknowledgements

[Acknowledgements go here.]

Contents

Declaration	ii
Abstract	iii
Publications	iv
Acknowledgements	vi
1 Introduction	1
1.1 Stellar probes of past mergers	2
1.1.1 Chemical tagging	3
1.1.2 Clustering in integrals-of-motion space	4
1.1.3 Stellar ages	4
1.1.4 Contamination from the in-situ halo	5
1.2 The assembly history of the Milky Way	5
1.3 Data-driven Galactic Archaeology	9
1.3.1 Overview of current machine learning methods	10
1.4 Thesis outline	13
2 Machine learning for accreted vs. in-situ classification	15
2.1 Introduction	15
2.2 The ARTEMIS Simulations	19
2.2.1 Stellar parameters as features	22
2.2.2 Galaxy-specific features	23
2.3 Machine learning models	24
2.3.1 Performance metrics	25
2.3.2 Supervised ML models	27
2.4 Model comparison	34
2.4.1 Classification performance	34
2.4.2 Model comparison on physical diagnostics	36
2.4.3 The ML performance in separating components versus observa- tional cuts	41

2.4.4	Visualisation of accreted and in-situ structures with UMAP	45
2.5	Testing the models on the Auriga simulations	47
2.6	Conclusions	50
3	Optimized HDBSCAN clustering for reconstructing the merger history of the Milky Way: applications and limitations	54
3.1	Introduction	54
3.2	The Auriga Simulations	57
3.2.1	The simulations sample	59
3.3	The HDBSCAN methodology	62
3.3.1	Optimizing the HDBSCAN parameters with OPTUNA	62
3.3.2	Evaluation metrics	64
3.3.3	The 12-dimensional parameter (feature) space	65
3.4	Results of Clustering	66
3.4.1	Clustering in accreted-only stellar haloes	67
3.4.2	Clustering in accreted + in situ stellar haloes	76
3.4.3	The impact of stellar ages	82
3.5	Discussion	85
3.6	Conclusions	86
4	GalactiKit: reconstructing mergers from $z = 0$ debris using simulation-based inference in Auriga	88
4.1	Introduction	88
4.2	The Auriga simulations	91
4.2.1	Extracting mergers information	92
4.3	Simulation-based inference	95
4.3.1	Density estimation with Masked Autoregressive Flows	97
4.3.2	Data pre-processing and training implementation	101
4.4	Results	102
4.5	Discussion	110
4.6	Conclusions	113
5	A simulation-based inference of the Milky Way merger history	116
5.1	Introduction	116
5.2	Data	119
5.3	Inferring Merger Properties with Simulation-Based Inference	121
5.3.1	A calibration model for generating realistic training data	124
5.3.2	Data aggregation with Fishnets	128
5.3.3	Posterior density estimation and validation	128
5.4	The time of accretion and stellar mass of the MW disrupted progenitors .	131

5.4.1	The mass-metallicity relation of the MW progenitors	135
5.4.2	The growth of the accreted stellar mass and halo mass of the MW	138
5.5	Conclusions	143
6	Conclusions	145
A	Merger posterior distributions of MW accreted substructures	149

List of Figures

1.1	Assembly history of the Milky Way inferred from the Galactic globular cluster population using the E-Mosaics simulations (Pfeffer et al., 2018). Figure taken from Kruijssen et al. (2020).	8
1.2	Composite photometry of the SDSS <i>i</i> -band, <i>r</i> -band, and <i>g</i> -band for the galaxies in the ARTEMIS simulation suite (Font et al., 2020). Figure taken from Font et al. (2020).	11
2.1	PR-AUC scores for the precision-recall curves for the models obtained training the benchmark architecture using different combinations of features. The feature categories are: positions and kinematics (labelled 'kin'), [Fe/H] and [α /Fe] abundances ('chem'), ages, and <i>Gaia</i> magnitudes and colours ('phot').	28
2.2	Comparison between the precision-recall curves obtained training the benchmark architecture and shallower ANNs on the optimal set of features (see Section 2.3.2.2).	30
2.3	Precision and recall values at different classification thresholds for the models. The metrics were evaluated considering all the stars in the test dataset.	34
2.4	Distribution of the FPs in the test set in the [α /Fe] – [Fe/H] plane (top row) and in the Toomre diagram (bottom row). Columns from left to right correspond to the MLP, MLP+galaxy features, TML and XGBoost models, respectively. For each panel, the top and side sub-panels show the probability density function of the FP distributions (yellow) and of the accreted (grey) and in-situ (blue) training examples. For each model, we also show the FP fraction (f_{FP}) of the total number of stars in the test dataset. The metrics are evaluated at the fiducial threshold values listed in Table 2.3.	37
2.5	Same as in Fig. 2.4, but for the accreted stars mis-classified as in-situ, i.e., the false negatives.	37

2.6	Information gain values describing the importance of the input features used by the XGBoost model to distinguish between accreted and in-situ stars.	39
2.7	The distribution in the $E - L_z$ plane of the accreted sample retrieved by the MLP model, at different classification thresholds (left to right panels), for each of the galaxy in the test dataset (top to bottom rows corresponding to galaxies G29, G30, G34, and G42, respectively.). The completeness (R) of the retrieved sample is also reported. The distribution is colour-coded by the actual fraction of accreted stars as defined by the simulation label.	40
2.8	The $[\alpha/\text{Fe}] - [\text{Fe}/\text{H}]$ distribution, colour-coded by the fraction of accreted stars, f_{acc} for stars in the simulated Solar neighbourhoods. From top to bottom, the rows correspond to galaxies G29, G30, G34, and G42, respectively. In the columns, the accreted stars are defined by: 1) the simulation label; 2-4) the observational selection criteria; 5) the label predicted by the MLP model. In each panel, we show the actual (column 1) and predicted (columns 2-5) overall fractions of accreted stars in the simulated Solar neighbourhoods.	42
2.9	Same as in Fig. 2.8, but showing results for the distributions of accreted stars in the Toomre diagram.	43
2.10	Distribution of stars in the training (left panel) and test (central and right panel) datasets in the parameter space defined by UMAP to maximise the separation between the accreted and in-situ populations. Colours represent the fraction of accreted stars in each region of the plane, as defined by the simulation labels (left and central panels) and by the predictions of the MLP model (right panel).	46
2.11	UMAP projections of the distribution of accreted and in-situ examples in the training data, colour-coded by the value of the corresponding physical parameters from the optimal set of features, used as input to the UMAP model. All values are normalised.	46
2.12	Precision (P) and recall (R) at different classification thresholds for the benchmark model, MLP, MLP+ galaxy features, TMP and XGBoost, trained on stars from six Auriga galaxies. The metrics are evaluated considering all stars in galaxies Au6 and Au21.	49
2.13	Comparison of the classification performance of models trained on ARTEMIS and tested on Auriga, and vice versa (orange lines). Included are also the performance of models trained and tested on data from the same simulation (purple lines).	51

3.1	Optimal HDBSCAN parameter configurations found by the Optuna search. Panels (a) and (b) show the results of optimizations using the V-measure and DBCV metrics, respectively. In both panels, each symbol represents a different galaxy. The top sub-panels show the configurations coloured by the fraction of particles classified as noise, while the bottom sub-panels show the same configurations coloured by the number of clusters in the resulting partition.	68
3.2	Density distribution of accreted star particles in the $E - L_z$ plane. The top row shows the ground-truth distribution, where particles are coloured by their progenitor group (PeakMassIndex). The middle and bottom rows show the HDBSCAN clustering results when optimized with the V-measure and DBCV metrics, respectively. In these lower panels, identified clusters are coloured according to their majority progenitor group, with noise particles shown in grey. The overplotted scatter points highlight the highest-density region of each cluster. The fraction of noise points is also included in each panel. At the bottom of each column, a legend shows the colours associated with the merger events reported in Table 3.1.	70
3.3	Precision scores for the clusters identified in accreted-only haloes by the HDBSCAN models optimized with the V-measure and DBCV index metrics. For each cluster, the precision is calculated with respect to the (dominant) progenitor with the highest number of cluster members, and plotted against the lookback infall time (top) and stellar mass at infall of the progenitor (bottom). The clusters are represented as triangles if the dominant progenitor is a stellar stream at $z = 0$ and by circles if a phase-mixed structure. The size of the symbols is proportional to the number of members in the cluster, and the colour scheme follows the one in Fig. 3.2 and indicates the dominant progenitor of each cluster.	71
3.4	Recall scores for the clusters identified in accreted-only haloes by the HDBSCAN models optimized with the V-measure and DBCV index metrics. The symbols, colours and sizes are the same as described in Fig. 3.3, but now for the recall scores.	73
3.5	Total variance values for the progenitor galaxies appearing as dominant progenitors in the HDBSCAN clusters.	74
3.6	Mahalanobis distance between the 12-D debris distributions of the progenitor with the largest number of particles classified as noise and the other progenitors of the galaxies. The symbol and colours of the progenitors are the same one used in Fig. 3.3.	75

3.7	Merger trees for the galaxies in the sample as reconstructed by the V-measure (middle) and DBCV (bottom) optimized HDBSCAN models for the accreted-only haloes. Top panel shows the full merger trees. Each colour represents a different progenitor galaxy. The thickness of each line is proportional to the number of accreted star particles from that progenitor. Dashed lines indicate progenitors that contribute $< 1\%$ to the overall accreted population of the galaxy at $z = 0$	76
3.8	Density distributions in the $E - L_z$ plane for the accreted stars (similar to Fig. 3.2), but now including the in situ stars which are shown in grey. The clustering results from the optimized models are shown in the middle panel for the V-measure and the bottom panel for DBCV. The noise distributions are not shown, however we report the fraction of points classified as noise in each panel.	77
3.9	Mahalanobis distance between the particle distributions of the progenitor galaxies and the in situ population. Colours are associated to progenitors using the same colour scheme as in Fig. 3.2.	79
3.10	Precision scores for the clusters identified by the HDBSCAN models optimized with the V-measure and DBCV index metrics for the accreted + in situ haloes. Symbols and colours are the same as in Fig. 3.3, for the accreted-only haloes. Crosses indicate clusters that are dominated by in situ particles and their associated “lookback infall time” is the 90 th percentile of the age distribution of the cluster.	80
3.11	Recall scores for the clusters identified in accreted + in situ haloes by the HDBSCAN models optimized with the V-measure and DBCV index metrics. Symbols, sizes and colours are the same as in Fig. 3.10.	81
3.12	Merger trees for the galaxies in the sample as reconstructed by the V-measure (middle) and DBCV (bottom) optimized HDBSCAN models when contamination from in situ stars is included. Top panel shows the full merger trees. The correspondence between lines and progenitors is the same as described in Fig. 3.7 for the accreted-only case. The style and thickness of the lines reflects the new fraction of accreted stars from each progenitor after the velocity selection cut is applied.	82
3.13	Density distribution in the $E - L_z$ plane for the stars in Au27 colour-coded by their HDBSCAN label. The clustering analysis is repeated excluding the age of the star particles from both the accreted-only and accreted + in-situ datasets. The plot format follows Figs. 3.2 and 3.8.	84

4.1	Accreted star particles located within R_{200} in the Au21 simulation at $z = 0$. The top row shows the integrals-of-motion distribution of the particles, while the $[\text{Fe}/\text{H}] - [\alpha/\text{Fe}]$ plane is reported in the bottom. For the plots on the left, the same colour is given to particles coming from each of the 30 progenitor galaxies contributing with at least 100 star particles. Only the <code>PeakMassIndex</code> of two progenitor galaxies is explicitly shown in the figure, while ellipsis are reported for the colours corresponding to the remaining progenitors. In the plots in the middle column, star particles are colour-coded differently if their progenitors are disrupted in the main halo (black) or are orbiting subhalos within R_{200} (gray). In the rightmost column, the plots show the distribution of star particles colour-coded based on the lookback time at which their associated progenitor galaxy crossed R_{200} for the first time.	94
4.2	Overview of the GalactiKit methodology. The Auriga simulations were first split into a training and test sets. For both cases, the accreted stars within R_{200} of the main halo were associated to the corresponding progenitor galaxy creating a set of merger-debris pairs $\{\mathbf{x}, \boldsymbol{\theta}\}$; this was then pre-processed to be used as input to the MAF models approximating $p(\boldsymbol{\theta} \mathbf{x})$. The quality of the estimation was assessed through summary statistics comparing the inferred values to the actual ones for the properties of mergers in the test. To ensure robust performance assessment, the procedure was repeated while systematically leaving out one Auriga simulation for testing at a time. The images of the Auriga galaxies are taken from Grand et al. (2017).	98
4.3	Merger parameters inferred by SBI models trained on different combinations of $z = 0$ debris properties: (a) E , L (top), (b) $[\text{Fe}/\text{H}]$, $[\alpha/\text{Fe}]$ (middle) and (c) E , L , $[\text{Fe}/\text{H}]$ and $[\alpha/\text{Fe}]$ (bottom). Each point represents the median value of the parameter samples, while the error bar shows the extent of the 34 th – 68 th percentile range of the distribution. A unique colour is associated to merger events associated to a specific galaxy, for which case the SBI models used for inference were trained on the mergers from the remaining galaxies in the suite.	104
4.4	Same as Fig. 4.3c, but mergers are colour-coded by the median value of the total energy (a, upper panel) and total angular momentum (b, bottom panel) of the corresponding debris distribution.	105
4.5	Same as Fig. 4.3c, but mergers are colour-coded by the median value of the $[\text{Fe}/\text{H}]$ (a, upper panel) and $[\alpha/\text{Fe}]$ (b, bottom panel) distributions of the corresponding debris.	107

5.1	Distribution of the orbital properties (E and L_z) of the accreted stars in the solar neighbourhood of the MW. The accreted stars are grouped by the proposed progenitor galaxy in which they formed, according to the criteria of Horta et al. (2023), and are plotted on top of the overall distribution of stars in the APOGEE sample from which they were selected.	122
5.2	Chemical abundance ratios ($[Mg/Fe]$ and $[Fe/H]$) distribution for the same samples of stars plotted in Fig. 5.1.	123
5.3	Relations between data, $\mathbf{x}$, and parameters, $\boldsymbol{\theta}$ for the the model, $\mathcal{M}$, describing the relation between merger and debris properties, and the calibration model, $\mathcal{C}$, describing the difference between simulated and observed stellar properties. The parameters referring to each model are indicated by dashed boxes.	125
5.4	Chemo-dynamical properties of the accreted star particles from the Auriga simulations used to train the posterior estimator (blue) and of the stars in the MW parent sample obtained from the APOGEE and Gaia surveys (red). The distribution of the MW stars belonging to the accreted substructures considered is also shown (pink). The distributions are normalised by the total amount of samples.	127
5.5	Outline of the pipeline used for the modelling of the posterior distribution of the properties of merger events, $\boldsymbol{\theta}$, from the observed $z = 0$ chemo-dynamical properties of their debris, $\hat{\mathbf{x}}$	130
5.6	Predicted vs. true parameters for the validation set. The top panels show the median of the predicted posterior (y-axis) against the true value (x-axis). Error bars represent the 16th – 84th percentile range. The bottom panels show the normalized residuals, with the green shaded region indicating the ± 1 deviation range.	130
5.7	Validation of posterior calibration using the TARP method. The plot shows the empirical coverage probability (y-axis) versus the predicted credibility level (x-axis). The dashed black line represents a perfect estimator.	131
5.8	Comparison of the lookback infall time (top) and stellar mass (bottom) of the MW progenitors as predicted in this work (bands) and through other studies in the literature (scatter points and bars). Scatter points represent single-value estimates, whereas bars indicate estimates quoting a confidence interval. The estimates from the literature for the Sequoia are all represented with the same marker and colour corresponding to the <i>N20</i> sample, but should be compared to the predictions from the <i>M19</i> and <i>K19</i> samples, too.	132

5.9	The mass–metallicity relation for the disrupted progenitors of the MW. Each progenitor is represented as a scatter point consistently with the scheme used in Fig. 5.8. The point is located at the median of the observed [Fe/H] debris distribution and the median of the inferred posterior distribution of the progenitor stellar mass. The extent of the errorbars shows the 16th – 84th percentile range in each dimension. In the background, the distribution of the Auriga mergers is shown in mass–metallicity bins colour-coded by their average redshift of infall. The continuous black line represents the median metallicity in each mass bin, while the dashed line shows the MZR inferred for the MW progenitors by Naidu et al. (2020).	137
5.10	Mass–metallicity relation for the disrupted satellites of the MW as in Fig. 5.9. The dashed lines represent the empirical MZR calculated with equation (5.2) for $z = 0$ (top line), $z = 1$, $z = 2$, $z = 3$ and $z = 4$ (top to bottom), respectively. The MZR at $z = 0$ is the same as the Kirby et al. (2013) relation.	138
5.11	Cumulative accreted stellar mass distribution for the MW through cosmic time with (top) and without (bottom) the Sagittarius merger. This calculation only accounts for 1 Sequoia sample (<i>K19</i>). The plot also reports measurements of the stellar mass of the MW halo, excluding the Magellanic Clouds, obtained through isochrone fitting (Deason et al., 2019) and density modeling (Mackereth & Bovy, 2020) of RGB stars.	141
5.12	Distribution of the MW dark matter halo mass estimates as a function of redshift, inferred from the posterior samples of the progenitor halo masses and corresponding halo-mass ratios with the MW at infall. The black line indicates the median MW halo mass at a given redshift, while the grey shaded area shows the 16th – 84th percentile range. For context, the 2σ coloured contours show the joint posterior distribution of the halo mass and infall redshift for each progenitor.	142
A.1		150
A.2		151
A.3		152
A.4		153
A.5		154
A.6		155
A.7		156
A.8		157
A.9		158
A.10		159

List of Tables

2.1	Sample of disc-like galaxies in ARTEMIS selected based on their co-rotation parameter, κ_{co} . These galaxies are separated into two datasets used for training and test the performance of the ML models, respectively. The columns are: 1) galaxy label; 2) fraction of accreted stellar component (defined as the mass fraction of accreted star particles over the total stellar mass (in-situ + accreted); 3) co-rotational parameter; 4) the total stellar mass; 5) half stellar mass radius (defined as the radius enclosing 50% of the total stellar mass); 6) maximum circular velocity; 7) average [Fe/H] abundance; 8) average $[\alpha/\text{Fe}]$ abundance, where α is tracked by Mg abundance. Apart for the fraction of accreted stars, all quantities are computed within 30 kpc from the centre of the MW-mass galaxy.	21
2.2	PR-AUC scores for the MLP models trained on accreted and in-situ examples from the galaxy in the left-most column, and tested on the galaxies listed in the top row. Where the galaxy label is the same, stars in the validation dataset were considered.	31
2.3	Fiducial classification threshold values for the models. Each value is associated to the highest F1-score calculated based on the precision and recall values on the validation dataset.	35
2.4	Comparison of the precision and recall values evaluated on the test dataset at the fiducial thresholds for the ML models.	35
2.5	Comparison of the purity (P) and completeness (R) of the sample of accreted stars retrieved by using observational selection cuts (top three rows) and by the MLP model (bottom three rows) evaluated at different thresholds, in the simulated Solar neighbourhoods of galaxies G29, G30, G34 and G42.	41

3.1	Merger events contributing at least 1% of the accreted stars bound to the main halo at $z = 0$ for the galaxies in the sample. From the leftmost to the rightmost, the columns describe: (i) the index of the progenitor in the merger tree of the galaxy; (ii) the lookback infall time, i.e., the first time the progenitor crossed R_{200} of its host; (iii) the stellar mass of the progenitor at the time of infall; (iv) the fraction of star particles bound to the main halo at $z = 0$ which were accreted from the progenitor; (v) the present-day configuration of the progenitor debris as classified by Riley et al. (2025).	61
4.1	Quantitative comparison between the merger parameter estimates inferred through the posterior models trained on different combinations of debris information. From left to right, the columns report: (i) the inferred merger parameter; (ii) the root-mean-squared error (RMSE) between the actual and inferred merger parameter; (iii) the fraction of mergers whose actual parameters are within the 34 th –68 th percentile range of the samples drawn from the posterior model; (iv) the mean-relative uncertainty (MRU) for the 34 th –68 th percentile range; (v) and (vi) are the same as (iii) and (iv) but considering the 5 th –95 th percentile range as confidence interval.	109
5.1	Lookback infall time and stellar mass predictions of the MW progenitors. From left to right, the columns indicate: the name of the progenitor accreted substructure (i), the lookback infall time and stellar mass predictions provided by GalactiKit (ii, iv) and studies from the literature (iii, v). References: a) Gallart et al. (2019), b) Bonaca et al. (2020), c) Kruijssen et al. (2020), d) Hasselquist et al. (2021), e) Montalbán et al. (2021), f) Naidu et al. (2022), g) González-Koda et al. (2025), h) Woody et al. (2025), i) Kepley et al. (2007), j/K19) Koppelman et al. (2019), k) Ruiz-Lara et al. (2022b), l) Horta et al. (2021), m) Malhan et al. (2021), n) de Boer et al. (2015), o/M19) Myeong et al. (2019), p) Dodd et al. (2025), q) Helmi et al. (2018), r) Mackereth et al. (2019a), s) Vincenzo et al. (2019), t) Mackereth & Bovy (2020), u) Feuillet et al. (2020), v) Callingham et al. (2022), w) Han et al. (2022), x) Lane et al. (2023), y) Helmi & White (1999), z) Horta et al. (2025), aa) Niederste-Ostholt et al. (2012), bb) McConnachie (2012), cc) Koppelman et al. (2019).	133
5.2	Fraction of progenitor stellar mass to MW total accreted stellar mass. From left to right, the columns indicate the median (50 th), 84 th and 16 th percentile of the mass fractions MW progenitors estimated from 1,000 sampled realisations of the Galaxy assembly history.	139

Appendix A

Merger posterior distributions of MW accreted substructures

This appendix supplements Chapter 5. The posterior distributions of the lookback infall time, stellar and halo mass, and halo mass merger ratio of the MW accreted satellites were estimated from the present-day chemodynamics of their debris using an ensemble of Masked Autoregressive Flows following the GalactiKit methodology. In Chapter 5, samples from these distributions were used to compute summary statistics, providing a quantitative description of the properties of each merger event and, overall, of the MW assembly history.

In this section, the posterior sample distributions for the accreted substructures are reported in individual figures: GES (Fig. A.1), Helmi streams (Fig. A.2), Heracles (Fig. A.3), I’itoi (Fig. A.4), LMS-1/Wukong (Fig. A.5), Sagittarius (Fig. A.6), and Thamnos (Fig. A.10). For Sequoia, we provide three distributions conditioned on debris selected with different criteria: *K19* (Fig. A.7), *M19* (Fig. A.8), and *N20* (Fig. A.9). Each distribution is presented as a corner plot of the inferred galaxy merger properties, with one-dimensional density distributions showing the median and the 16th–84th percentile range. Samples are coloured consistently with the substructure colour scheme adopted in the analysis.

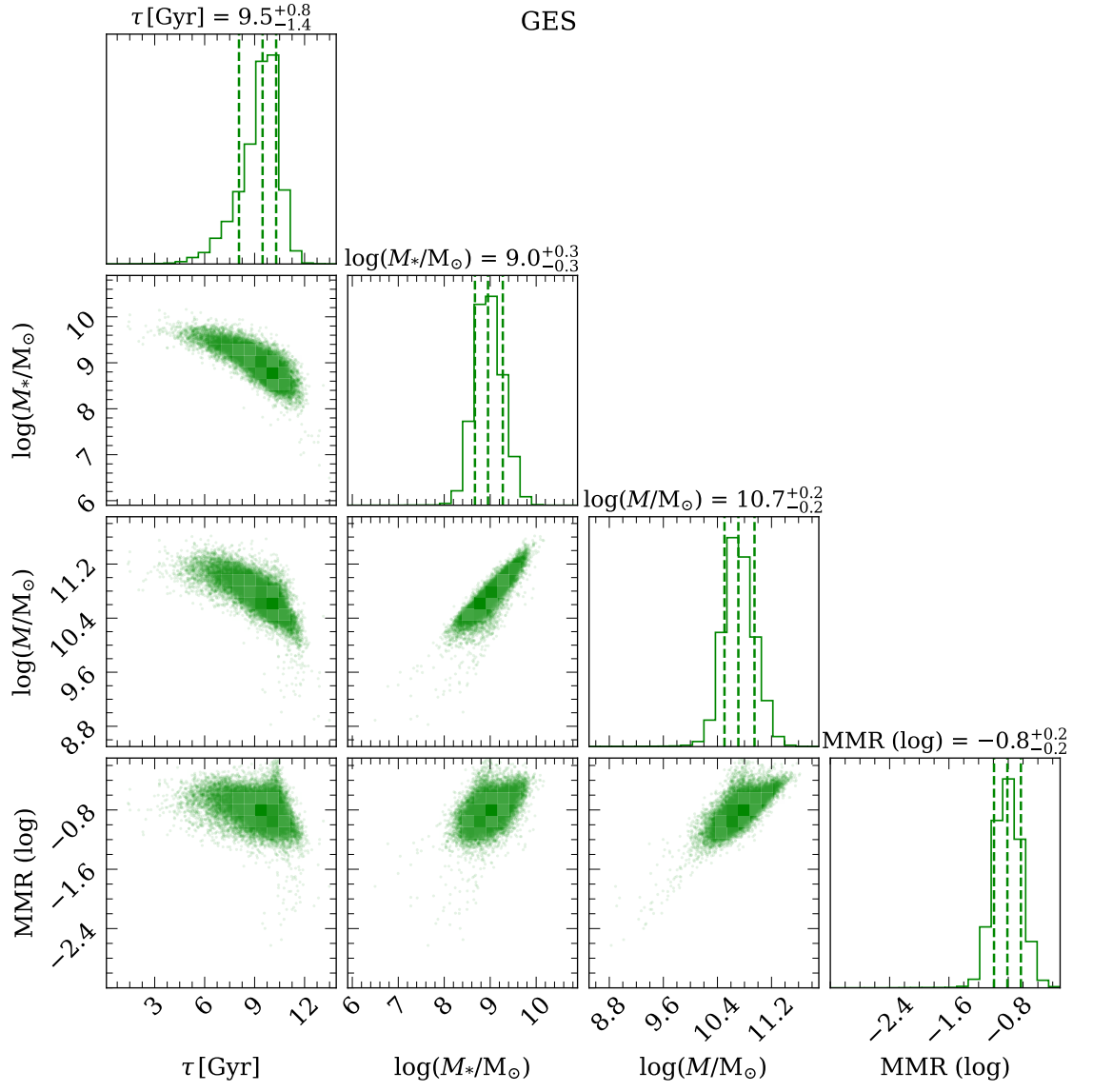


Figure A.1

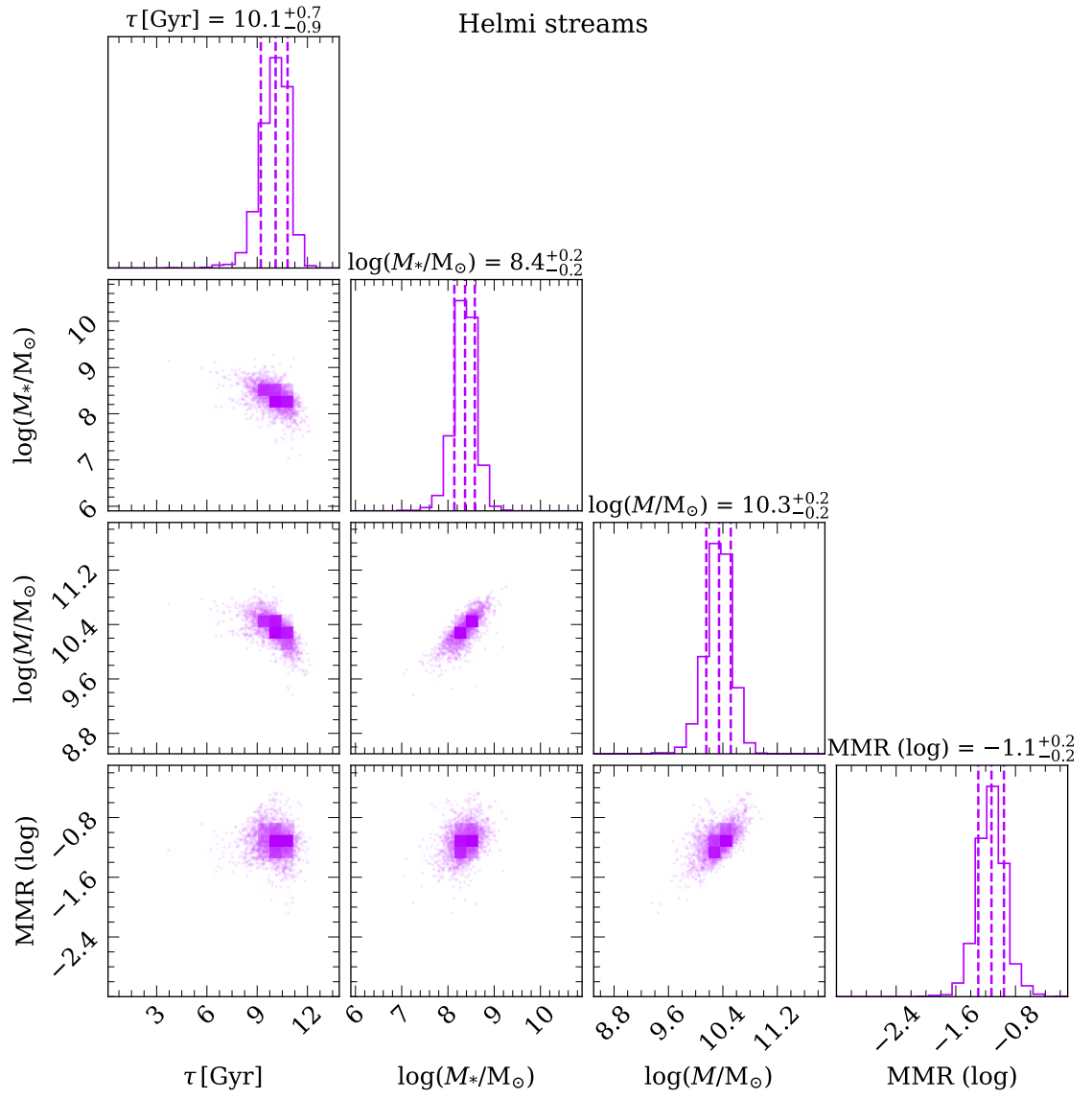


Figure A.2

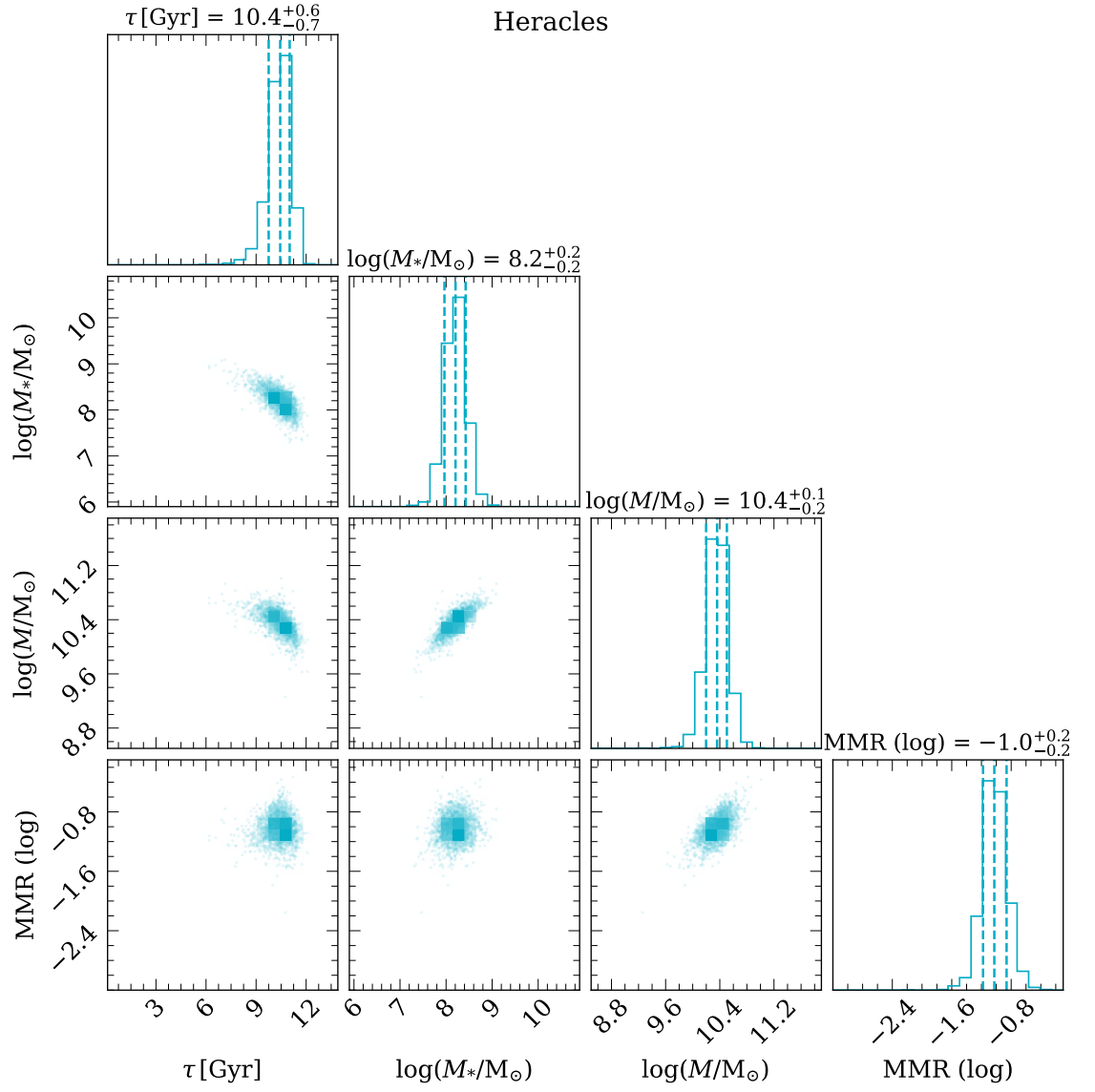


Figure A.3

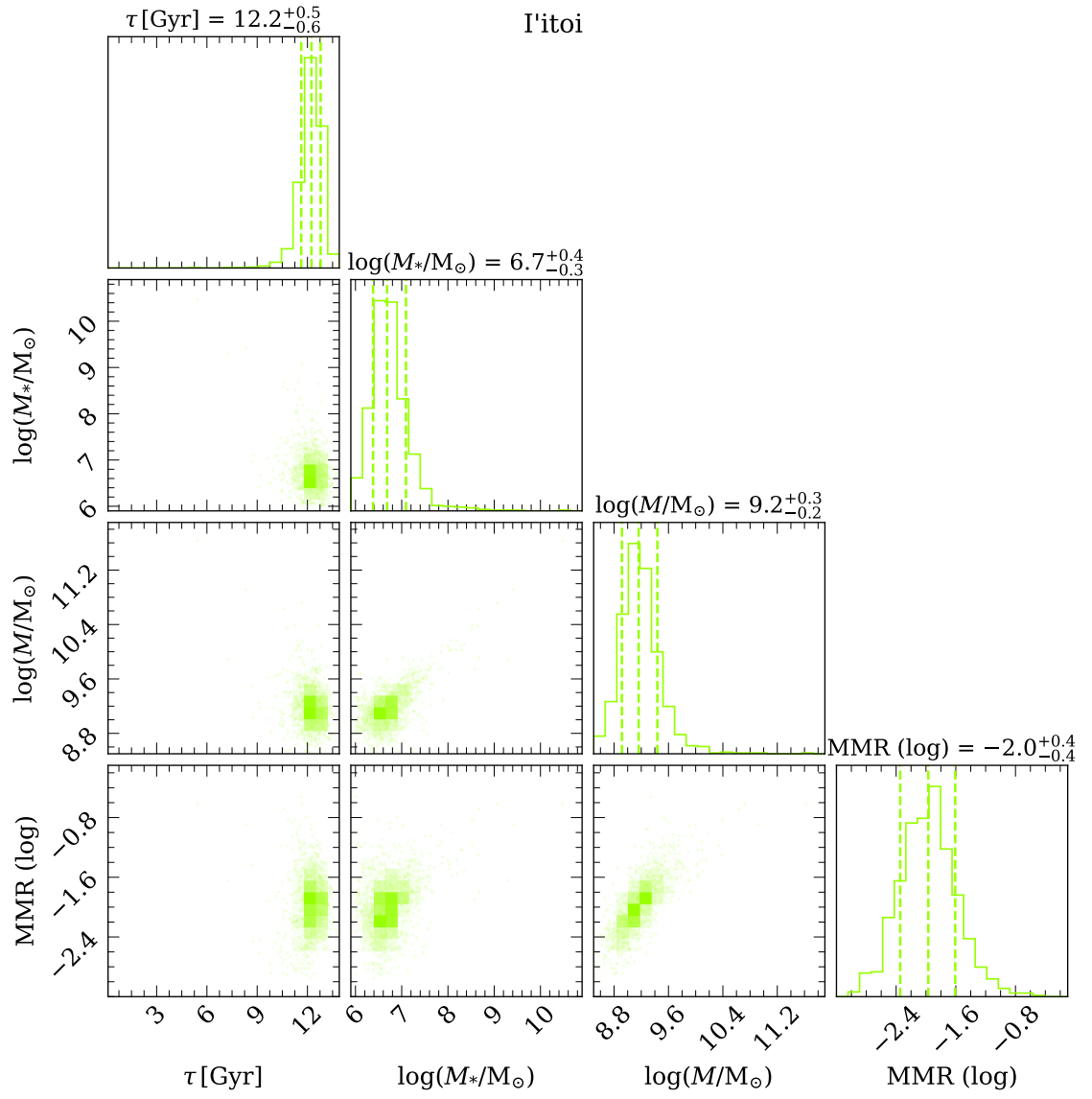


Figure A.4

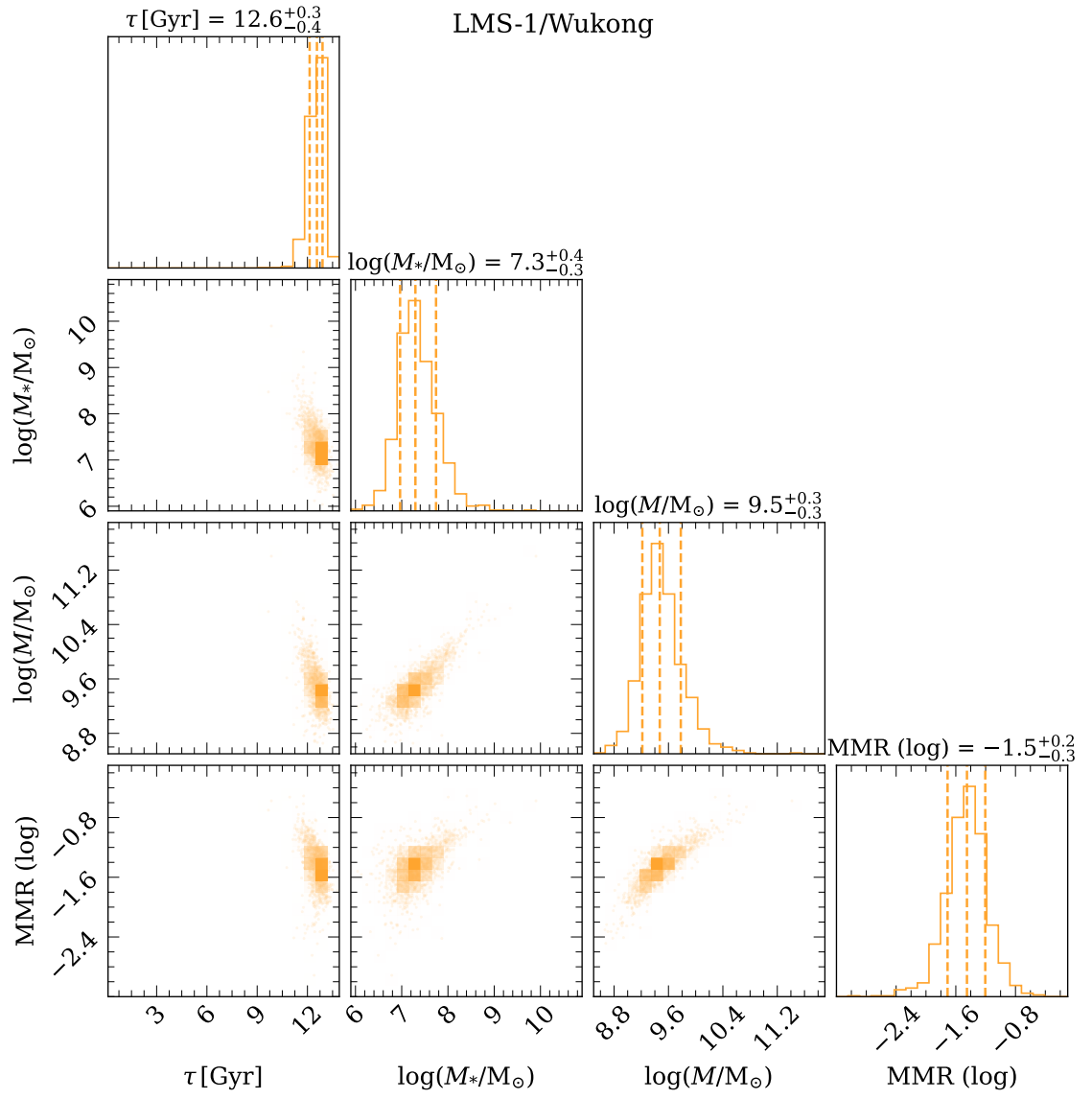


Figure A.5

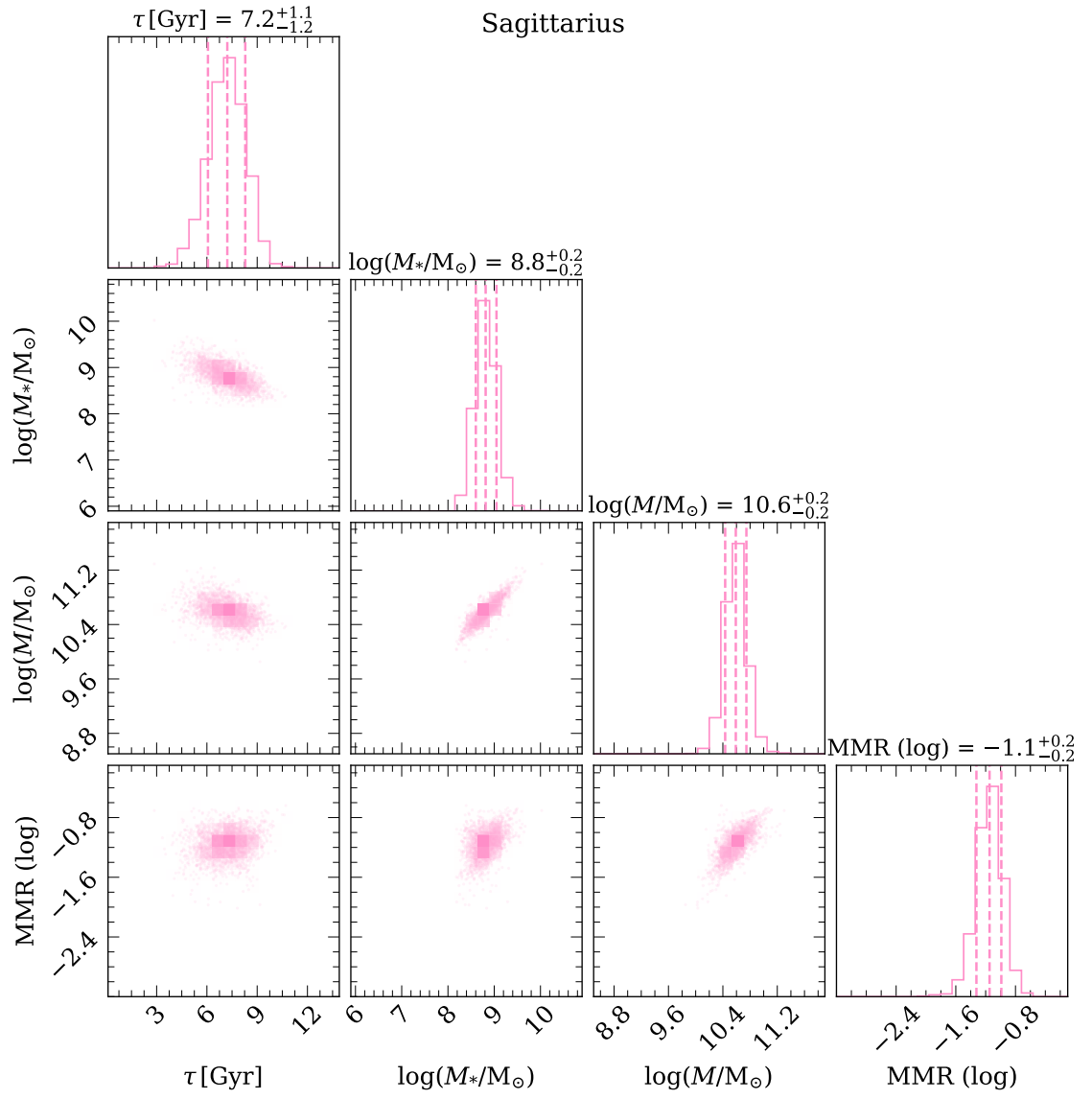


Figure A.6

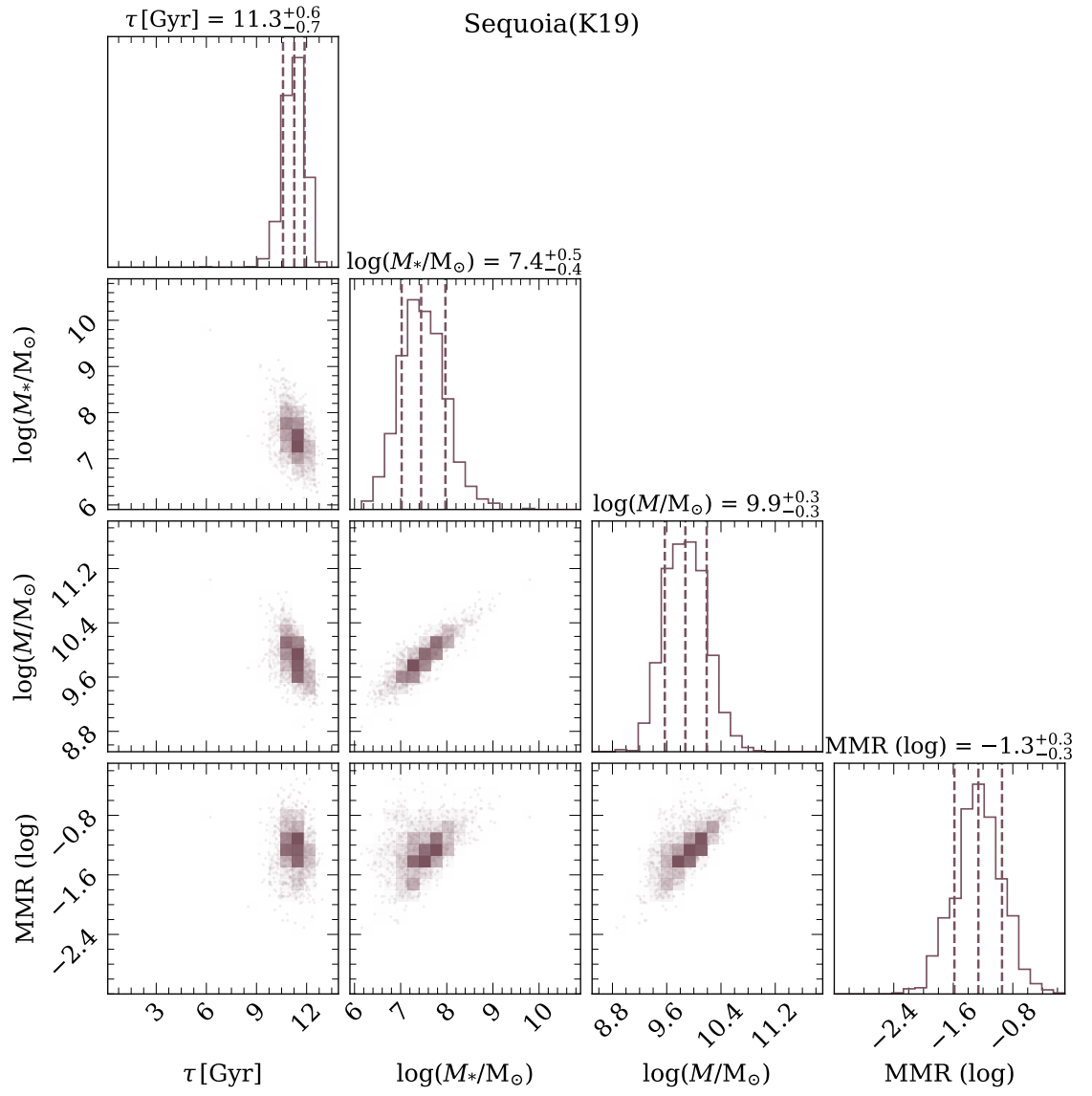


Figure A.7

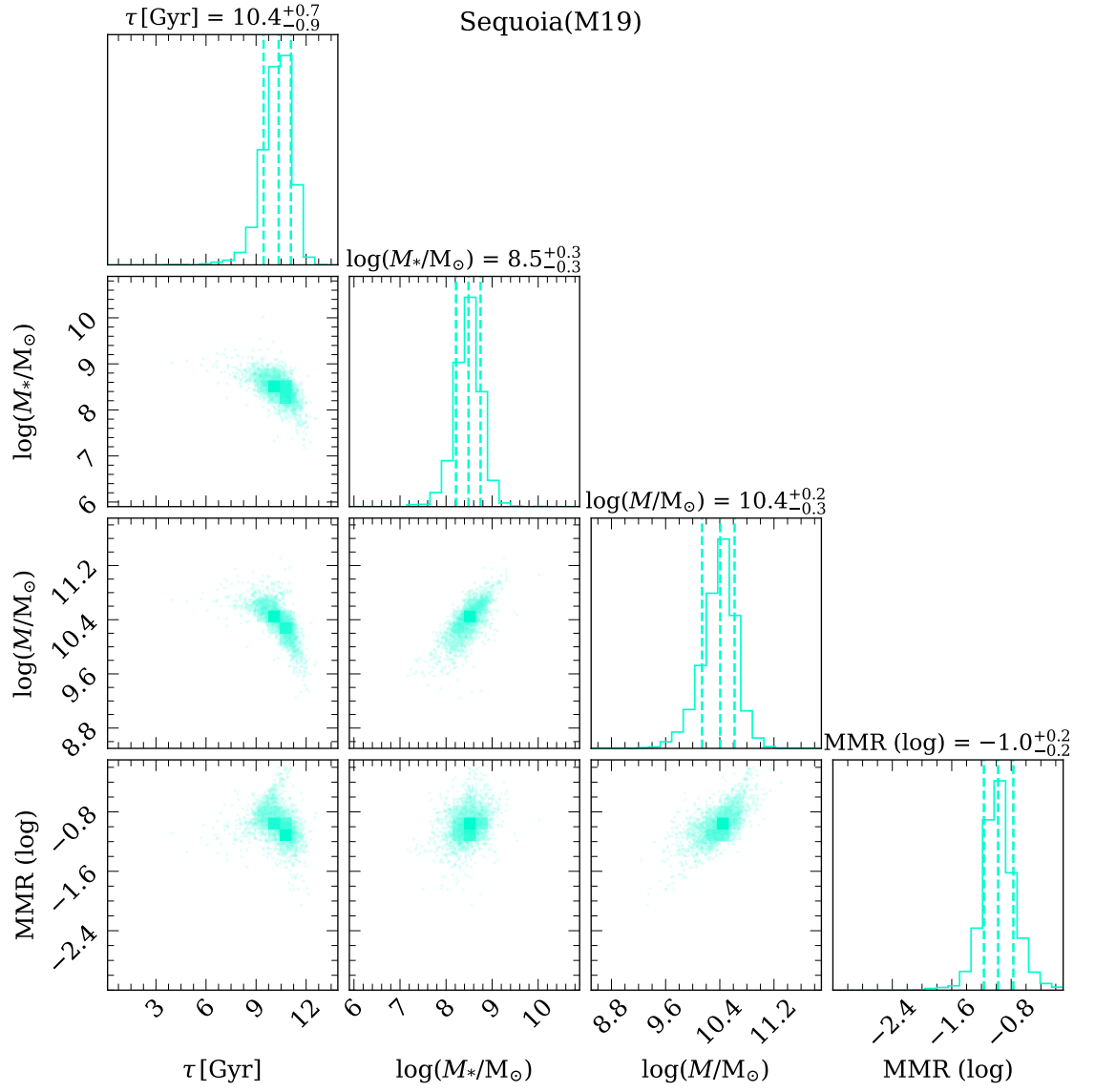


Figure A.8

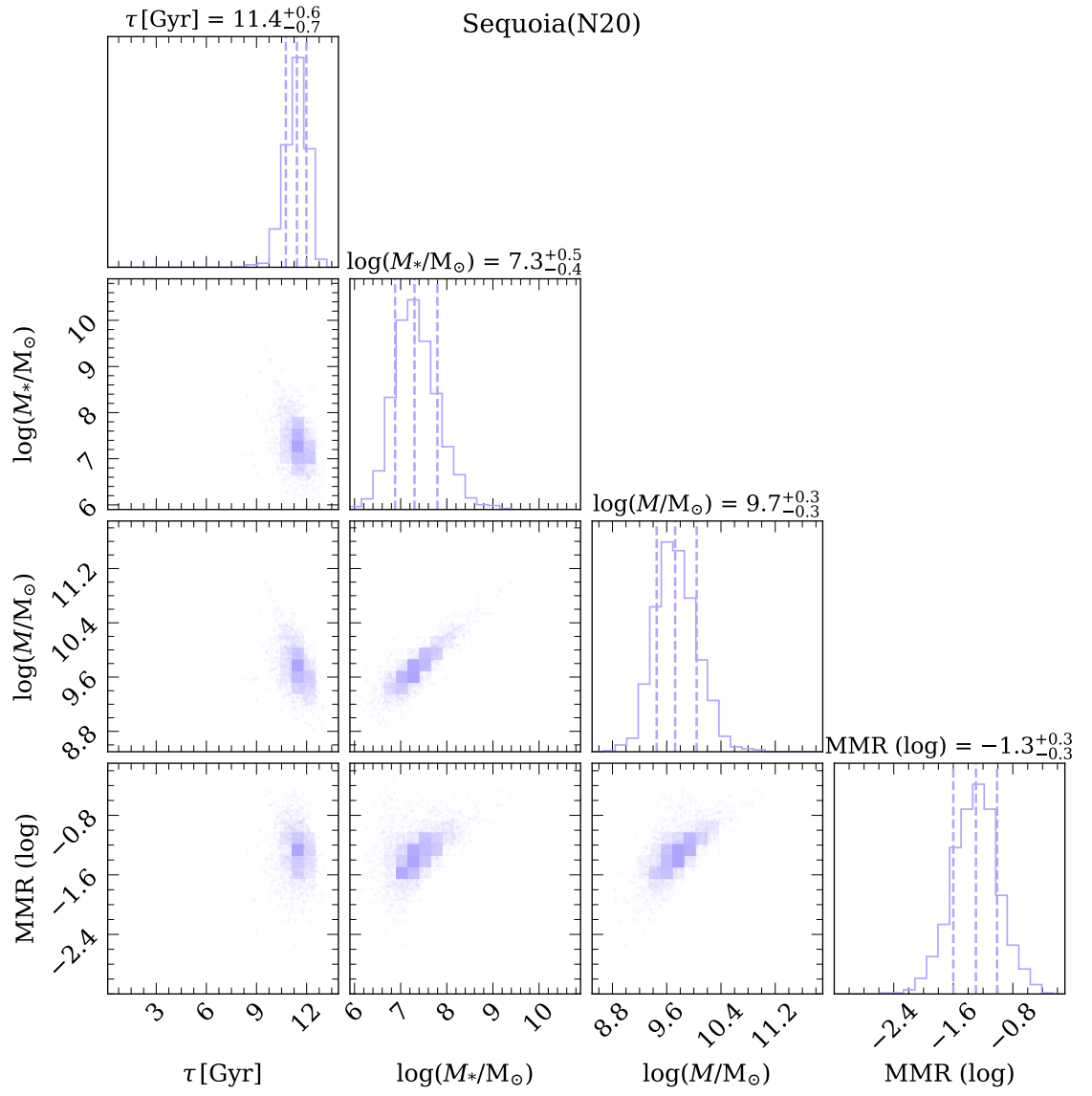


Figure A.9

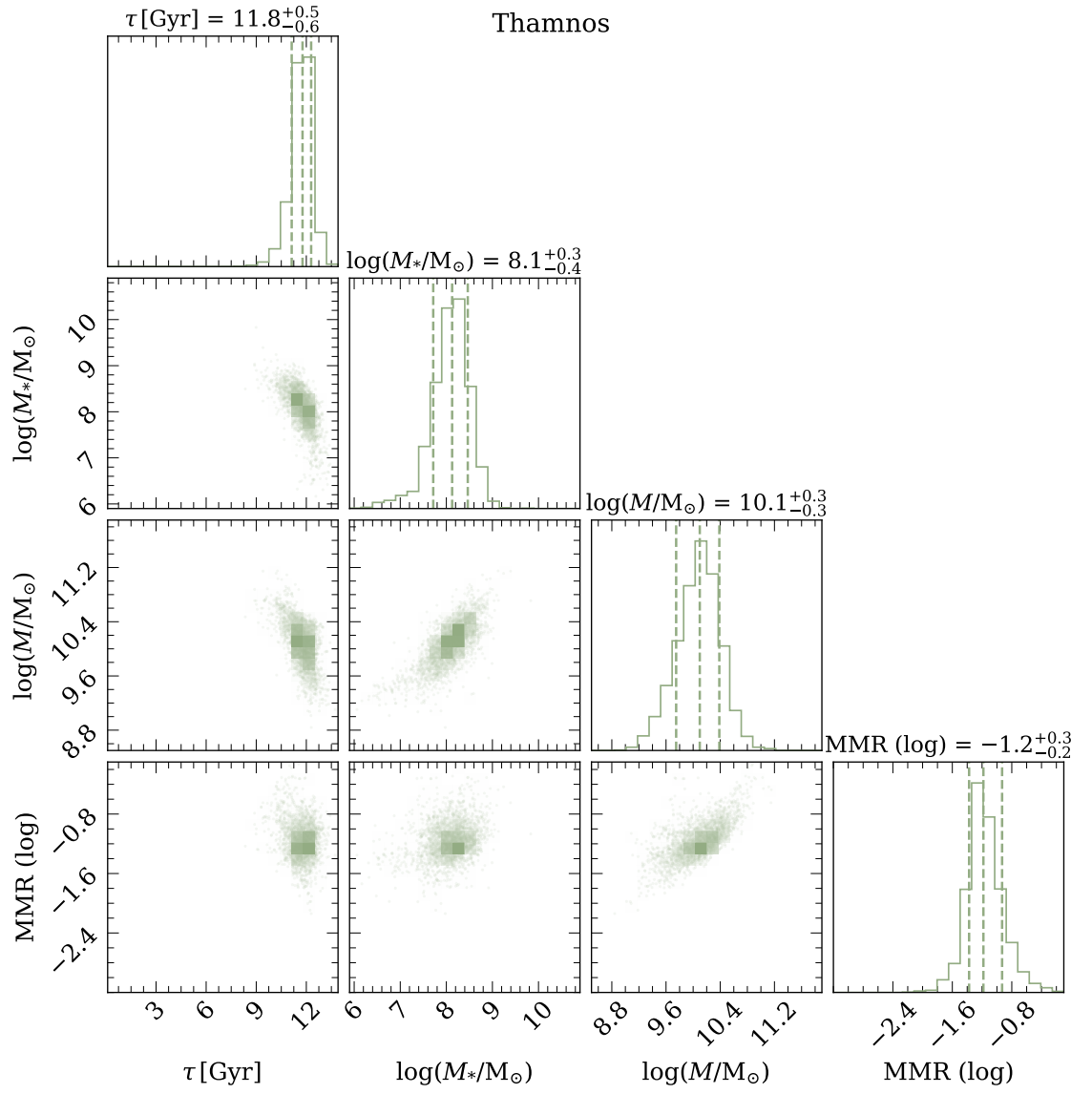


Figure A.10

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